Model Card – PeriCALM Patterns 3.0 (by PeriGen Inc.)

Model Details

- Detection of fetal heart rate (FHR) and uterine pressure (UP) patterns from electronic fetal monitoring (EFM) data
- Digital signal preprocessing and feature extraction
- LSTM-based neural network using TensorFlow
- Post-processing for event generation
- Executes in near real-time for clinical utility
- FDA cleared class II medical device (January 2025, version 2.0 was cleared in 2005)

Intended Use

- PeriCALM Patterns is intended for use to provide additional secondary information as an adjunct to qualified clinical decision-making during antepartum or intrapartum obstetrical monitoring at ≥32 weeks gestation for annotation and summary of the fetal heart rate recording for baseline, accelerations and decelerations and the uterine pressure recording for contractions.
- Uninterpretable FHR signal is identified.
- Uses fetal heart and maternal contraction tracing data collected from standard fetal monitors
- Integrated into PeriGen Inc. software (Vigilance)

Training Data

- The training dataset comprised 1853 hours from 342 tracings
- The training dataset considered a wide range of conditions including tracings with artifact, rare patterns, a range of sizes of accelerations and decelerations and from the antepartum and intrapartum periods of labor.
- Each FHR event (acceleration, deceleration, baseline etc.) was annotated by an academic Obstetrician Gynecologist to establish the training ground truth.
- Tracings were annotated with a software tool to indicate the time intervals of baseline, accelerations, deceleration and uninterpretable signal. This tool was used for both training and evaluation.

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Evaluation Data

- 30 Preterm and 40 Term tracings were used for the study.
- Three Truthers were selected to represent experts. All three have attained the rank of full professor in Obstetrics and Gynecology.
- Six additional Obstetrician Gynecologists were selected for a group called Readers. The group covers a range of obstetrical experience levels, practice in different types of hospitals, and geographical regions. Five were professors of Obstetrics and Gynecology; one was an Obstetrics and Gynecology practitioner.
- Truthers and Readers independently annotated all tracings with the same marking tool used for training.
- The Truther majority opinions were merged to establish the evaluation ground truth markings.
- We compared Patterns 3 and Reader performances with respect to this ground truth.

Metrics

- Separate analysis for preterm and term gestational ages
- Comparison of performances of Patterns 3 and Readers with respect to Truthers
- Evaluation metrics include:
  - accelerations and decelerations: mean and confidence intervals on estimates of sensitivity and specificity
  - baseline level estimates: limits of agreement (LoA) over 30-minute epochs

Bias & Ethical Considerations

- The model algorithm does not use any demographic information to inform its decisions.
- The only variable that was established for the clearance of the model was gestational age where some FHR parameters may not be equivalent. For that reason, the model was limited to GA >= 32 weeks where FHR interpretation was accepted by FDA as equivalent supported by statistical analysis.
- Current clinical practice of EFM pattern interpretation is standard for all maternal/fetal profiles. Race, age, BMI, height socio-economic background, etc. should play no role in interpretation. The characteristics of accelerations, decelerations, baseline and contractions are the same regardless of any of these potential bias categories.

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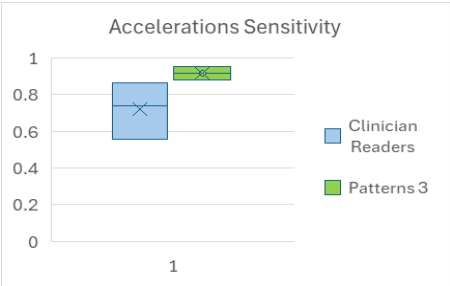
- To ensure that significant clinical factors did not affect the model output, the data was stratified by important clinical factors such as gestational age (GA), maternal BMI, 5-minute Apgar, umbilical cord arterial pH at birth (UA-pH), fetal birthweight and acquisition device. We saw no statistically significant change in performance for acceleration, deceleration and baseline detection across subgroups for each factor.

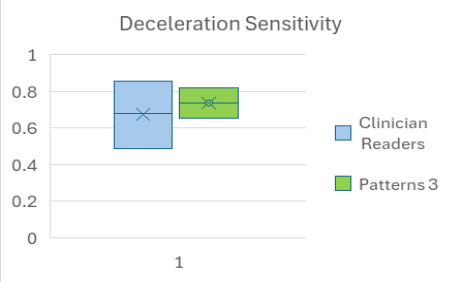
Quantitative Analyses

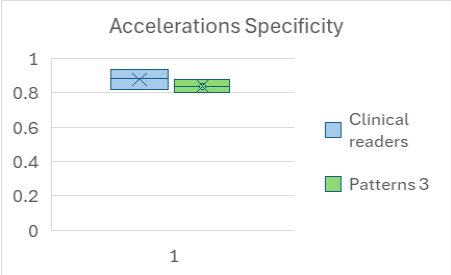
- The following tables and associated plots show performance metrics for term and preterm gestational age groups, comparing Patterns 3 algorithm detections and the clinical readers group markings
- For overall estimates of acceleration and deceleration sensitivity and specificity we show mean, lower and upper confidence intervals (LCI and UCI)
- For baselines we show the bias and LoA of the baseline level estimate (bpm)

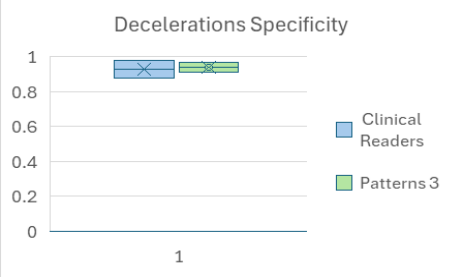
<u>For Term fetuses (GA >= 36 weeks):</u>

Term gestations		Clinical Readers			Patterns 3.0		
		Mean	LCI	UCI	Mean	LCI	UCI
Accelerations	Sensitivity	0.739	0.559	0.865	0.918	0.879	0.951
	Specificity	0.882	0.818	0.936	0.839	0.8	0.877
Decelerations	Sensitivity	0.678	0.488	0.856	0.737	0.652	0.817
	Specificity	0.928	0.874	0.976	0.94	0.912	0.964









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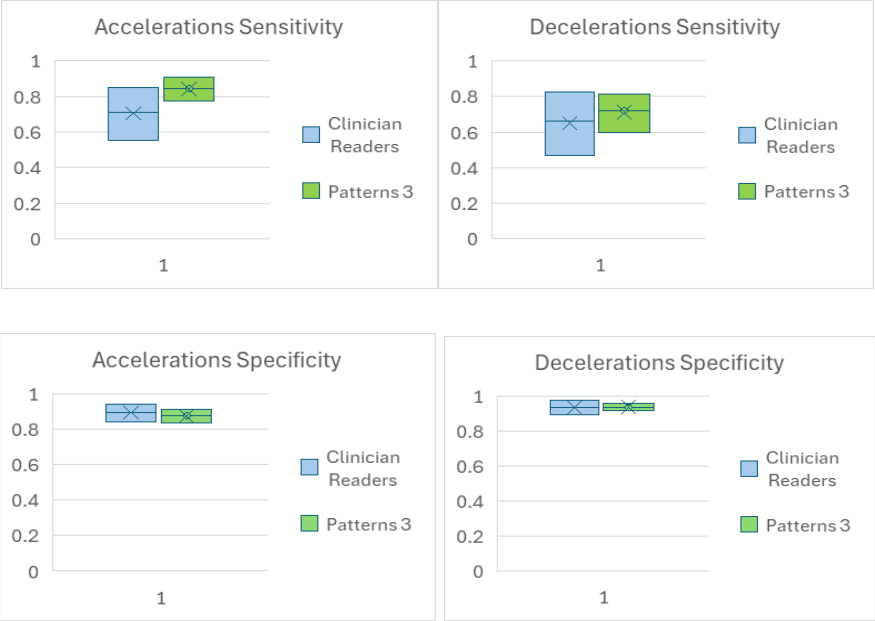
Term fetuses: Sensitivity and Specificity for accelerations and decelerations, clinician readers vs Patterns 3

- From the above chart for term fetuses, compared to Clinical readers, Patterns 3 has improved sensitivity (true positives) and a largely equivalent specificity (false positives).

For Preterm fetuses (GA >= 32 weeks and <36 weeks):

Preterm gestations		Clinical Readers			Patterns 3.0		
		Mean	LCI	UCI	Mean	LCI	UCI
Accelerations	Sensitivity	0.709	0.556	0.851	0.844	0.772	0.906
	Specificity	0.897	0.841	0.944	0.875	0.835	0.91
Decelerations	Sensitivity	0.659	0.47	0.824	0.72	0.597	0.814
	Specificity	0.937	0.893	0.976	0.934	0.918	0.962



Preterm fetuses: Sensitivity and Specificity for accelerations and decelerations, clinician readers vs Patterns 3

- From the above chart for preterm fetuses, compared to Clinical readers, Patterns 3 has improved sensitivity and a largely equivalent specificity with more robust estimates, as shown by the reduced height of the green bars vs blue bars.

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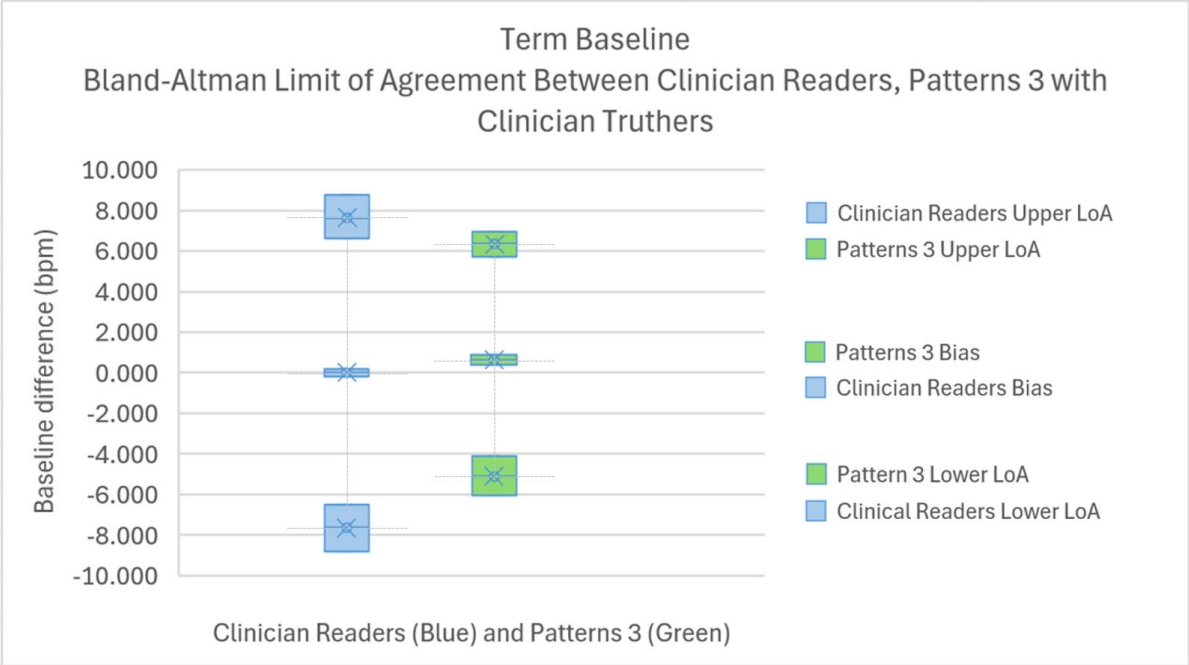
Baseline Term Fetuses:

Term gestations	Clinical Readers			Patterns 3.0		
	LoA Lower	Mean	LoA Upper	LoA Lower	Mean	LoA Upper
Lower baseline limits of agreement (LoA)	-8.821	-7.617	-6.508	-6.027	-5.105	-4.122
Bias	-0.198	0.006	0.201	0.630	0.384	0.881
Upper baseline limits of agreement (LoA)	6.602	7.629	8.781	5.715	6.365	6.957



From the above chart for term fetuses, compared to Clinical readers, Patterns 3 has a more robust estimate of baseline level, as shown by the narrower range of LoAs in the green bars vs blue bars, with largely equivalent bias (near 0 bpm).

- It should be noted that the smallest vertical spacing of typical tracing gridlines is 3 mm which equals an increment of 10 bpm. That is, resolution of baseline below increments of 5 bpm is beyond human visual acuity and clinicians do not use finer resolution in practice.

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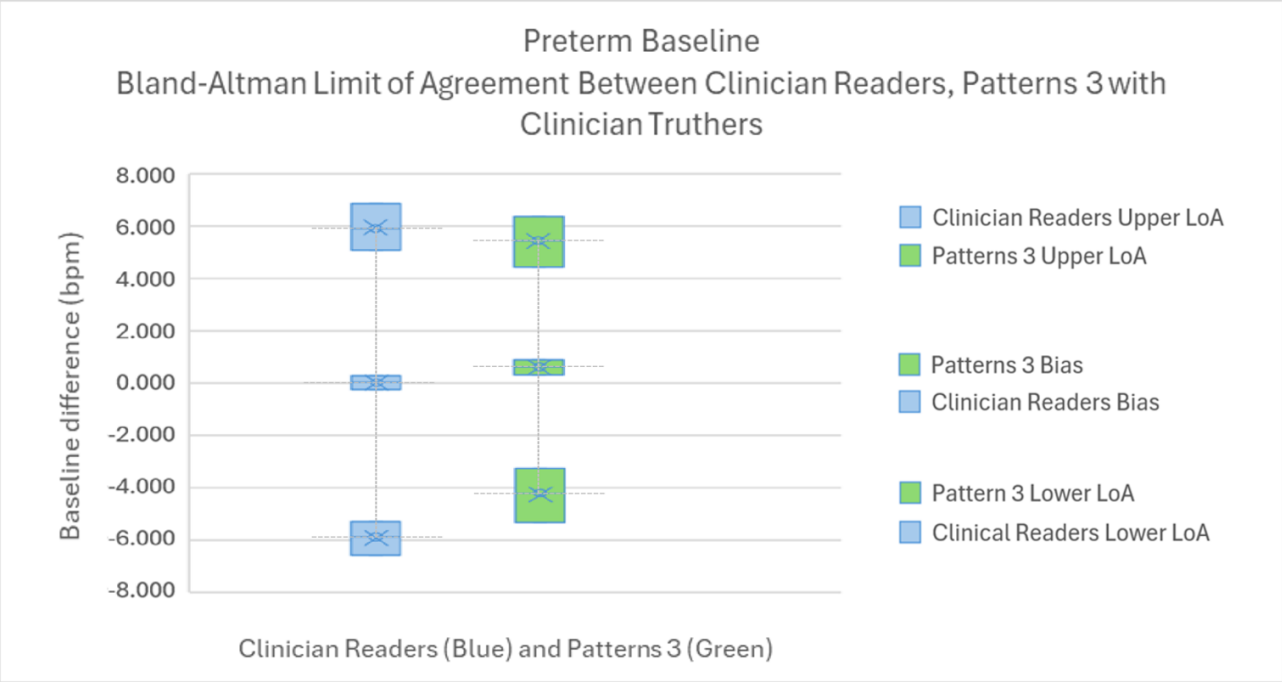
Baseline Preterm Fetuses:

Preterm gestations	Clinical Readers			Patterns 3.0		
	LoA Lower	Mean	LoA Upper	LoA Lower	Mean	LoA Upper
Lower baseline limits of agreement (LoA)	-6.587	-5.863	-5.277	-5.325	-4.197	-3.236
Bias	-0.220	0.028	0.296	0.324	0.629	0.917
Upper baseline limits of agreement (LoA)	5.084	5.919	6.885	4.458	5.455	6.395



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WARNING: Evaluation of FHR during labor and patient management decisions should not be based solely on PeriCALM Patterns annotations or summaries and should include inspection of the fetal monitor tracing and consideration of all pertinent clinical information.

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